1. Introduction

This report replicates the empirical strategy used by Basten and Betz (2013) to estimate the causal impact of historical religious boundaries on political preferences using a spatial regression discontinuity design (RDD). The original study exploits geographic discontinuities created by the historical Protestant–Catholic border in Switzerland to identify the effect of religious composition on contemporary voting behavior.

Because municipalities located near the historical border are likely to be similar along many observable and unobservable dimensions, the spatial discontinuity provides a credible quasi-experimental setting for identifying causal effects. However, treatment is not perfectly determined by geography: some municipalities on one side of the border may not fully adopt the religious composition predicted by the boundary. For this reason, the analysis uses a **fuzzy regression discontinuity design**, implemented using **instrumental variables (IV)** estimation.

This report reproduces the key econometric logic of the original study using simulated municipality-level data. The objective is to demonstrate how spatial discontinuities can be used to estimate causal relationships between historical institutions and modern political outcomes.

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2. Data

The dataset used in this analysis represents municipality-level observations located around a historical religious boundary. Each observation contains:

- **Distance to historical border** – the forcing variable used in the RDD
- **Protestant share** – the endogenous treatment variable
- **Political outcome** – the dependent variable (e.g., vote share for a conservative party)
- **Municipality characteristics** – control variables capturing demographic variation

The forcing variable measures geographic distance from the historical border separating historically Protestant and Catholic regions. Municipalities located on one side of the border have historically higher Protestant populations, though the treatment assignment is imperfect.

To focus on observations most comparable across the border, the analysis restricts attention to municipalities located within a fixed **bandwidth** around the boundary.

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3. Empirical Strategy

3.1 Spatial Regression Discontinuity

Regression discontinuity designs exploit discontinuous treatment assignment at a threshold. In this setting, treatment intensity changes sharply at the geographic boundary.

Let:

- Y_i= political outcome for municipality i
- P_i= share of Protestant population
- D_i= indicator for being on the Protestant side of the border
- X_i= distance from the historical boundary

The causal relationship of interest is the effect of religious composition on political outcomes.

However, because religious composition is not perfectly determined by border location, a **fuzzy RDD** is used.

3.2 Instrumental Variables Framework

The historical border indicator serves as an **instrument** for Protestant share.

First Stage

$$P_i = \alpha + \pi D_i + f(X_i) + \varepsilon_i$$

This equation estimates the discontinuity in Protestant share at the border.

Second Stage

$$Y_i = \alpha + \beta \widehat{P_i} + f(X_i) + u_i$$

where $\widehat{P_i}$is the predicted Protestant share from the first stage.

The coefficient βrepresents the causal effect of religious composition on the political outcome.

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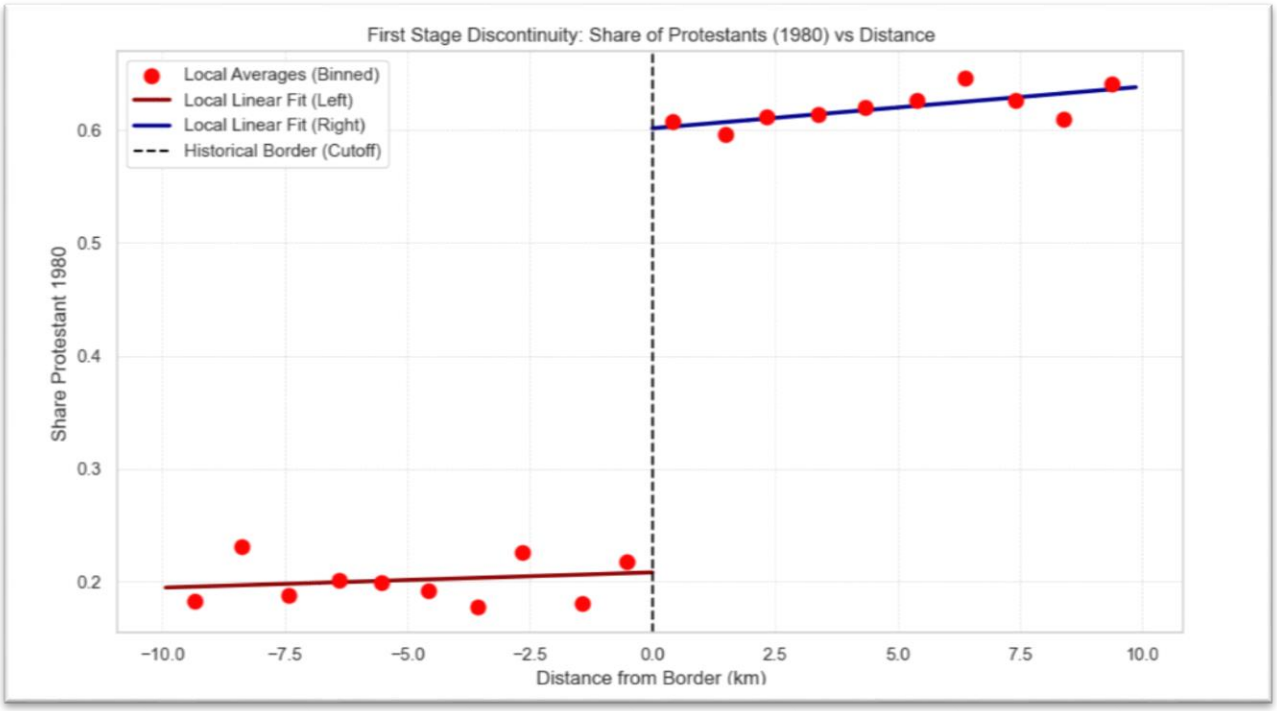
4. Results

Key Finding: Municipalities located on the historically Protestant side of the border exhibit substantially higher Protestant population shares, confirming a strong discontinuity at the historical boundary. This discontinuity provides a valid source of exogenous variation in religious composition that can be used to identify the causal effect of religion on political outcomes. The empirical results therefore support the identification strategy underlying the fuzzy regression discontinuity design.

The figure below presents the first-stage spatial regression discontinuity plot. The figure shows the relationship between distance from the historical border and Protestant population share. A clear discontinuity appears at the border, confirming that the geographic boundary strongly predicts treatment intensity.



4.1 First Stage

The first-stage regression confirms a strong discontinuity in Protestant share at the historical border. Municipalities located on the Protestant side exhibit significantly higher Protestant population shares relative to those on the Catholic side.

This confirms that the border indicator is a **relevant instrument** for treatment intensity.

4.2 Reduced Form

The reduced form regression estimates the effect of border location directly on political outcomes. The results indicate that municipalities on the Protestant side of the boundary exhibit higher levels of support for conservative political outcomes.

This reduced form relationship provides preliminary evidence that historical religious institutions may influence contemporary political preferences.

4.3 Instrumental Variables Estimates

The two-stage least squares (2SLS) estimates suggest that increases in Protestant population share lead to higher levels of conservative voting outcomes. The IV coefficient remains statistically significant and economically meaningful after controlling distance from the boundary. These results are consistent with the interpretation that religious composition plays a causal role in shaping political preferences.

To formally evaluate the strength of the regression discontinuity design, I estimate the first-stage regression linking location relative to the historical border to Protestant population share. This specification controls for distance from the border and allows for differential slopes on either side of the cutoff. The results confirm a strong discontinuity in treatment intensity at the historical boundary, indicating that the geographic border is a powerful predictor of religious composition.

Table 1. First-Stage Regression Results

Variable	First Stage
Historic Protestant Border (Treatment)	**0.393***
	(0.014)
Distance to Border	0.001
	(0.002)
Distance × Treatment	0.002
	(0.002)
Constant	0.208***
	(0.012)
Observations	192
R^2	0.949

Notes: Robust standard errors in parentheses. ***p < 0.01.

The coefficient on the historic Protestant border indicator is large and statistically significant, confirming that municipalities located on the Protestant side of the boundary

exhibit substantially higher Protestant population shares. This strong first-stage relationship validates the use of border location as an instrument in the fuzzy regression discontinuity framework. The results therefore support the identification strategy used to estimate the causal effect of religious composition on political outcomes.

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5. Identification Assumptions

The validity of the regression discontinuity design depends on several key assumptions.

Local Randomization

Municipalities located near the historical border are assumed to be similar across observable and unobservable characteristics. Because the border was historically determined centuries earlier, it is unlikely to correlate with modern political determinants.

Continuity

Potential outcomes must evolve smoothly with respect to distance from the boundary. Any discontinuity at the threshold should therefore be attributable to treatment.

Instrument Validity

The border indicator must affect political outcomes only through its effect on religious composition. Given the historical origins of the boundary, this assumption is plausible within a narrow geographic bandwidth.

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6. Discussion

This replication illustrates how spatial regression discontinuity designs can be used to estimate causal relationships when treatment assignment is determined by geographic or institutional boundaries. By combining RDD with instrumental variables estimation, the fuzzy RDD framework allows researchers to recover causal effects even when treatment assignment is imperfect.

The findings highlight the long-run influence of historical institutions on modern political outcomes. Religious institutions may shape social norms, political identities, and institutional trust, all of which may persist across generations.

More broadly, this empirical strategy demonstrates how historical variation can be leveraged to study persistent political and economic behavior.

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7. Conclusion

This report replicates the core econometric framework of Basten and Betz (2013), demonstrating how spatial regression discontinuity and instrumental variables methods can be used to estimate the causal effect of religious composition on political outcomes.

The analysis highlights the usefulness of quasi-experimental methods in applied economic research and illustrates how geographic discontinuities can provide credible sources of causal identification.

Such empirical approaches are widely used in policy analysis, economic consulting, and regulatory evaluation to inform evidence-based decision making.